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Advanced Cluster Management

Multicluster, Hosted Control Plane, GitOps, Resilience & DR

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Agenda

- ACM in context
- Multicluster management
- Hosted control plane
- GitOps integration
- OpenShift resilience
- Disaster recovery
- Summary

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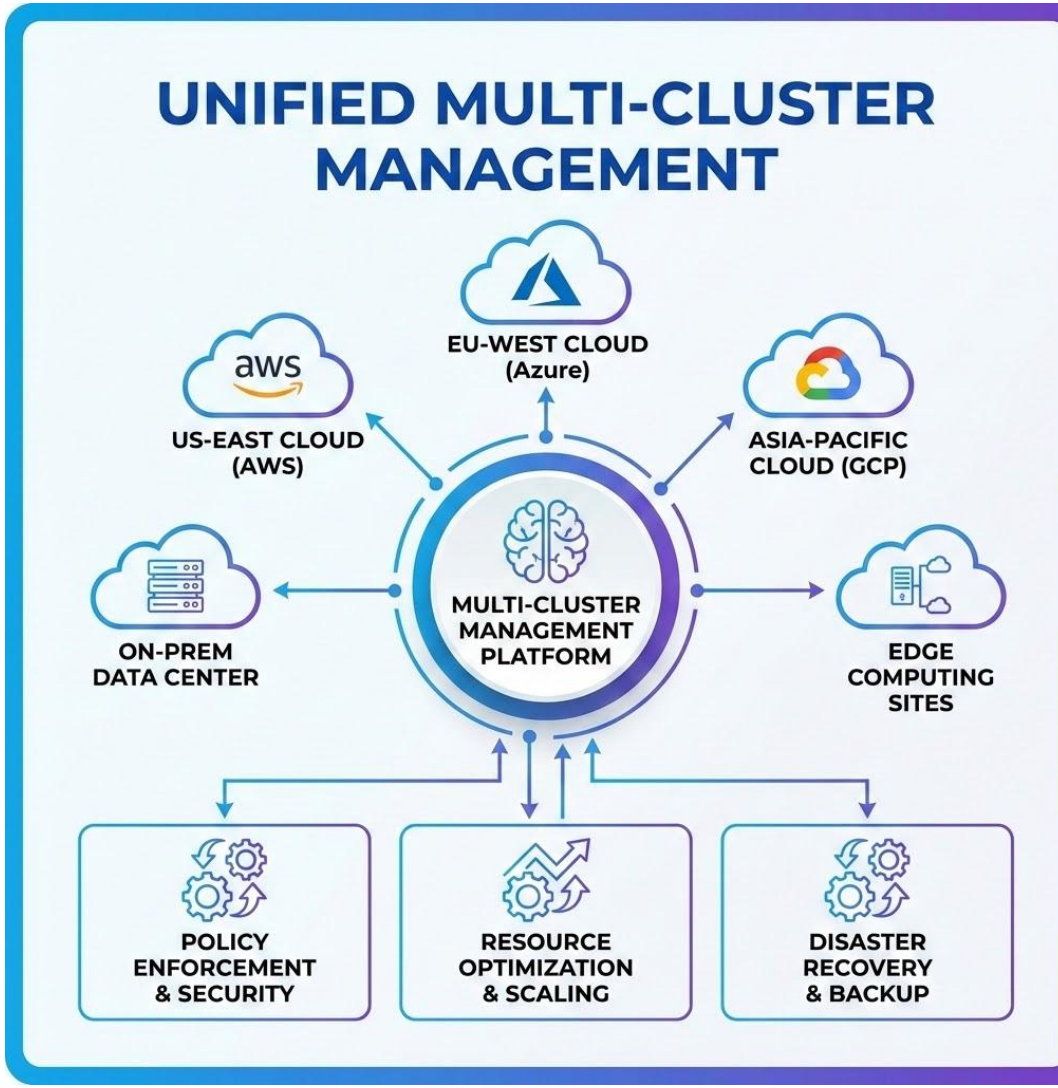
ACM in Context



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What Is ACM

The hub for your fleet

- Central management plane for multiple Kubernetes or OpenShift clusters
- Runs on a hub cluster; managed clusters are imported or created
- Policy-driven governance, compliance, and configuration at scale
- Application lifecycle across clusters (subscriptions, placement)
- Integrates with GitOps, observability, and hosted control plane patterns

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Multicluster Management

Fleet Operations

One pane of glass

- Import existing clusters or provision new ones from the hub (where supported)
- Cluster inventory, health, and upgrade visibility across the fleet
- Placements and placement rules: target workloads to cluster sets
- Labels, cluster sets, and resource binding for application distribution
- Consistent RBAC and authentication integration with corporate IdP

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Policies at Scale

Multicluster guardrails

- Policy controllers enforce standards (security, configuration, certificates)
- Report compliance status per cluster and namespace
- Remediate drift or alert operators when policies fail
- Integrate with organizational compliance frameworks
- Reduce configuration sprawl across dozens of clusters

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Hosted Control Plane

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Hosted Control Plane Overview

OpenShift and ACM

- Separates the Kubernetes control plane from worker nodes (logical separation)
- Enables dense multi-tenancy: many clusters with lower infrastructure footprint
- ACM can manage hosted-control-plane-based clusters as managed resources
- Useful for cloud, edge, and rapid cluster provisioning scenarios
- Operational model: hub manages lifecycle, upgrades, and placement of hosted clusters

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Why Hosted Control Plane + ACM

- Faster cluster creation and teardown for dev/test or edge
- Centralized upgrades and policy from the ACM hub
- Consistent security baseline across many small clusters
- Pairs with cluster pools and automation for large estates
- Reduces toil compared to traditional three-node control planes everywhere

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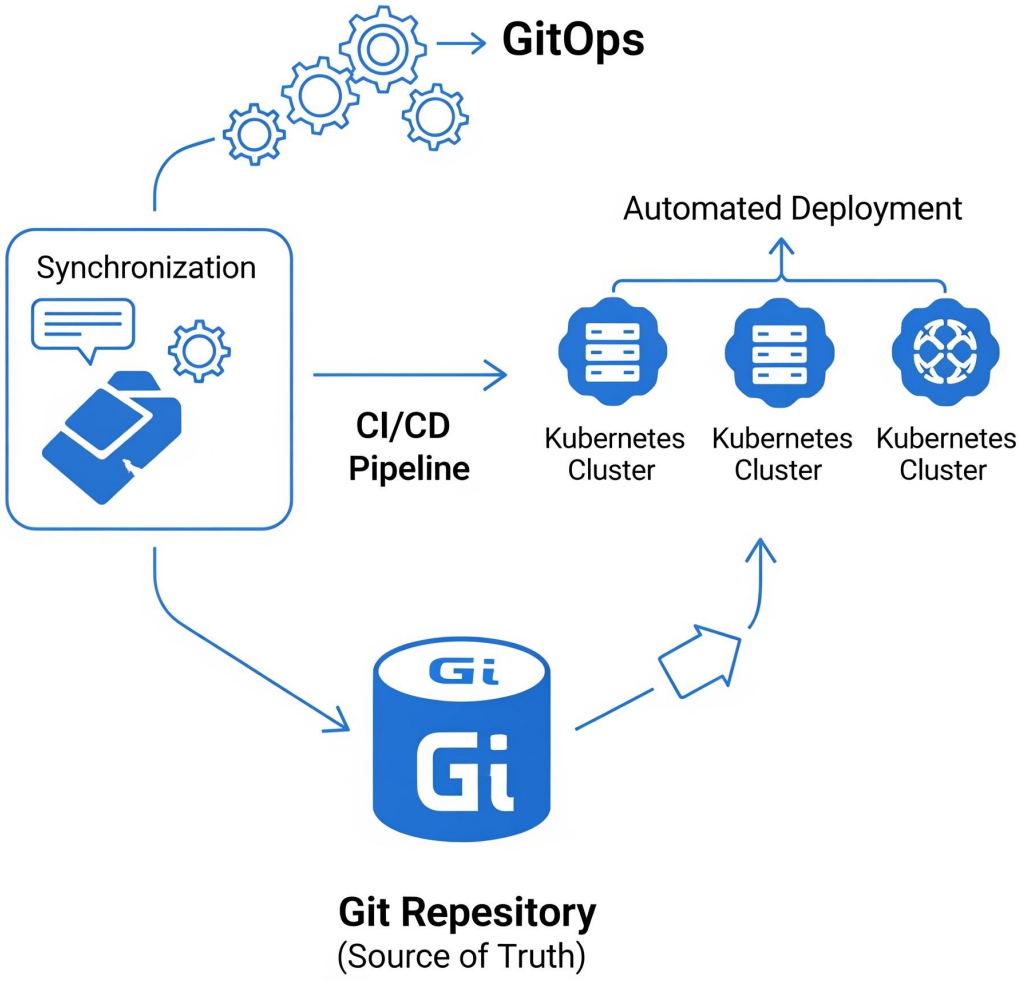
GitOps Integration



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GitOps with ACM

Argo CD and subscriptions

- ACM integrates with OpenShift GitOps (Argo CD) for multi-cluster application delivery
- Application and ApplicationSet patterns can target multiple managed clusters
- Subscriptions and channel resources propagate manifests from Git or Helm
- Align Git as source of truth with hub policies and placement
- Promote applications through dev → stage → prod across cluster sets

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The GitOps Revolution

Automated Management

VMS
NotebookLM

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GitOps – Operational Benefits

- Single workflow for platform config and app manifests
- Audit trail in Git; hub shows deployment status per cluster
- Rollback and promotion without logging into each cluster
- Policy ensures clusters meet prerequisites before apps land
- Supports hybrid and multi-cloud GitOps consistently

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OpenShift Resilience

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Resilience on OpenShift

What ACM helps monitor

- Highly available control plane and etcd (per cluster design)
- Machine health, node repair, and upgrade orchestration
- Observability stack: metrics, logs, alerts aggregated to the hub
- Network and storage resilience: CSI, Metro, Regional patterns
- Chaos testing and failure injection in non-production environments

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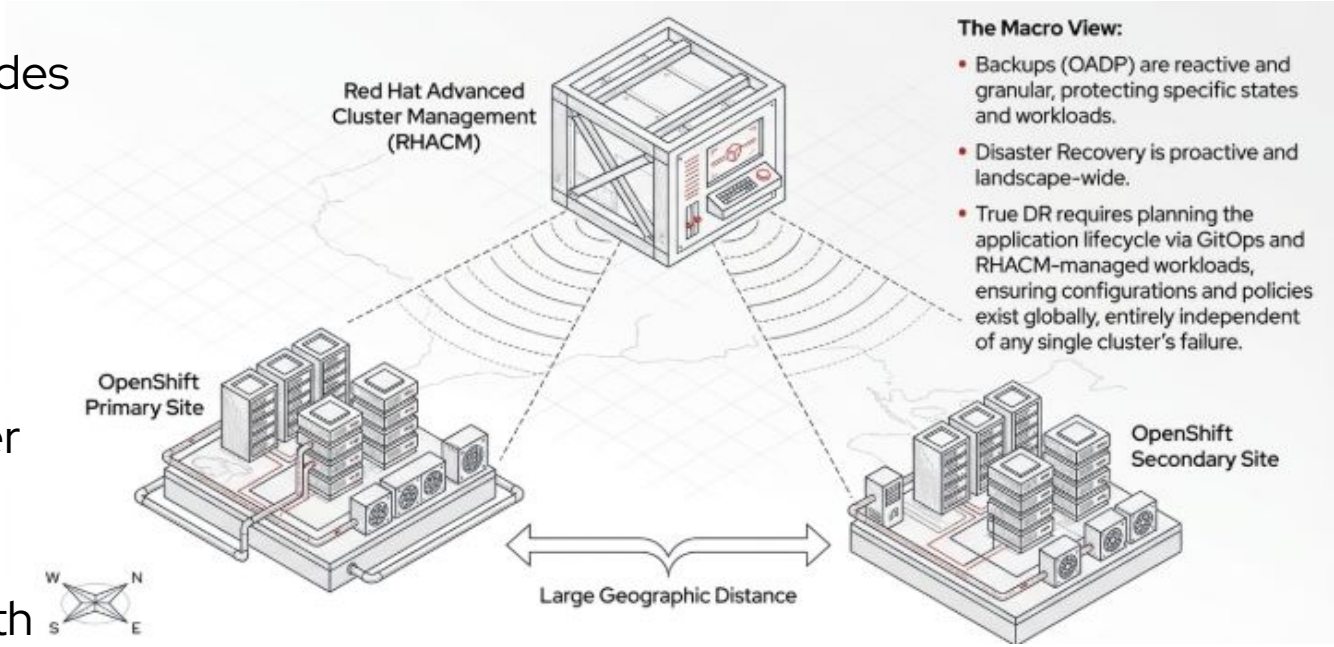
ACM and Resilience

- Detect unhealthy clusters or policy violations early
- Coordinate maintenance windows and upgrades across regions
- Backup and restore integrations can be orchestrated from policies
- Reduce blast radius by isolating workloads per cluster or cluster set
- Document RTO/RPO per tier and validate with game days



Disaster Recovery

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Disaster Recovery with ACM

High level

- ACM supports Regional DR patterns with ODF and VolSync (product-dependent)
- Hub can orchestrate failover policies and application placement to secondary sites
- Backup hub configuration and managed cluster metadata for hub recovery
- Combine application replication (OADP, storage) with ACM placement rules
- Test failover regularly; document runbooks for hub vs. spoke loss

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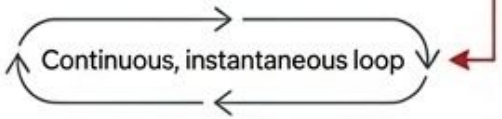
The Disaster Recovery Matrix

Metro-DR
Campus / City-Scale

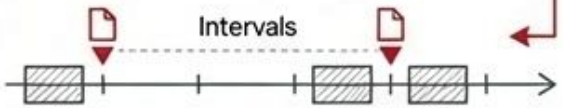
- Replication: Synchronous (Two-way mirroring).
- Network Tolerance: Strict < 10ms network round-trip time.
- Data Loss Window (RPO): Zero data loss.



- Underlying Technology: Red Hat OpenShift Data Foundation (ODF) / Ceph natively spanning data centers.

Regional-DR
Geographic / Continental

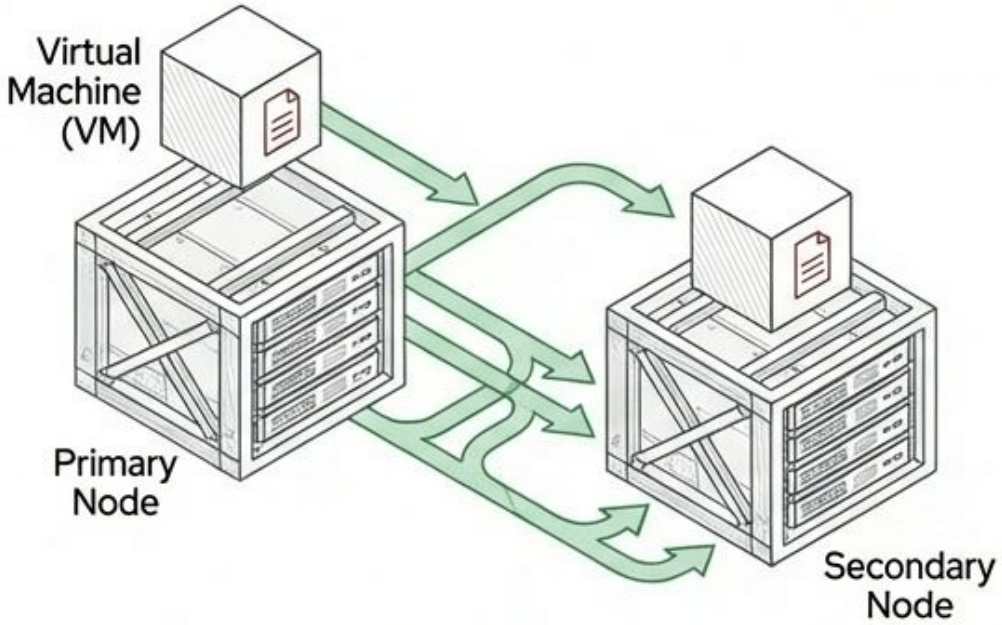
- Replication: Asynchronous (Snapshot-based syncing).
- Network Tolerance: High latency tolerant (Geographically dispersed).
- Data Loss Window (RPO): Proportional to the snapshot replication interval length.



- Underlying Technology: ODF VolumeReplicationClass / External object sync.

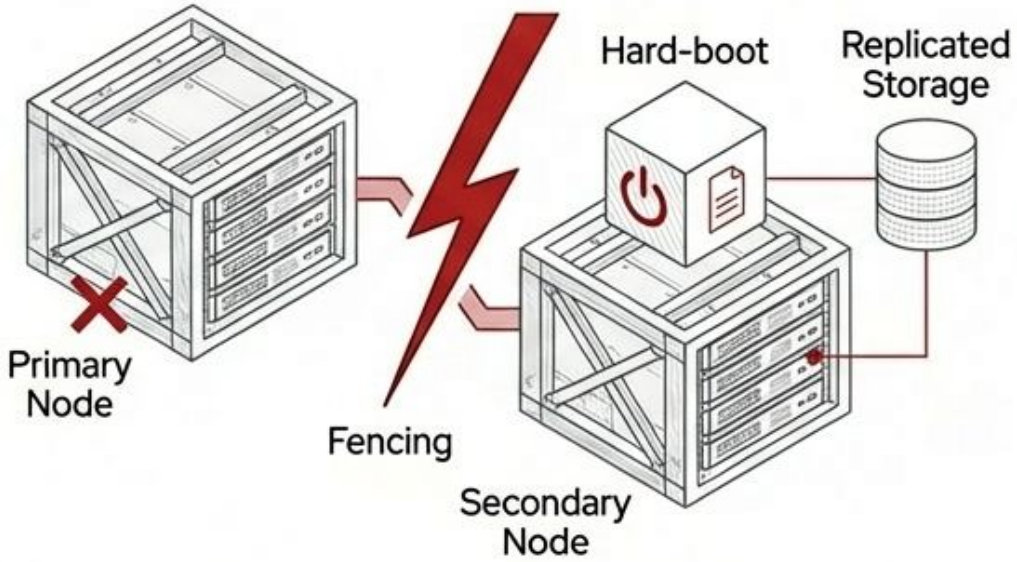
Relocate vs. Failover Workflows

Planned Relocation

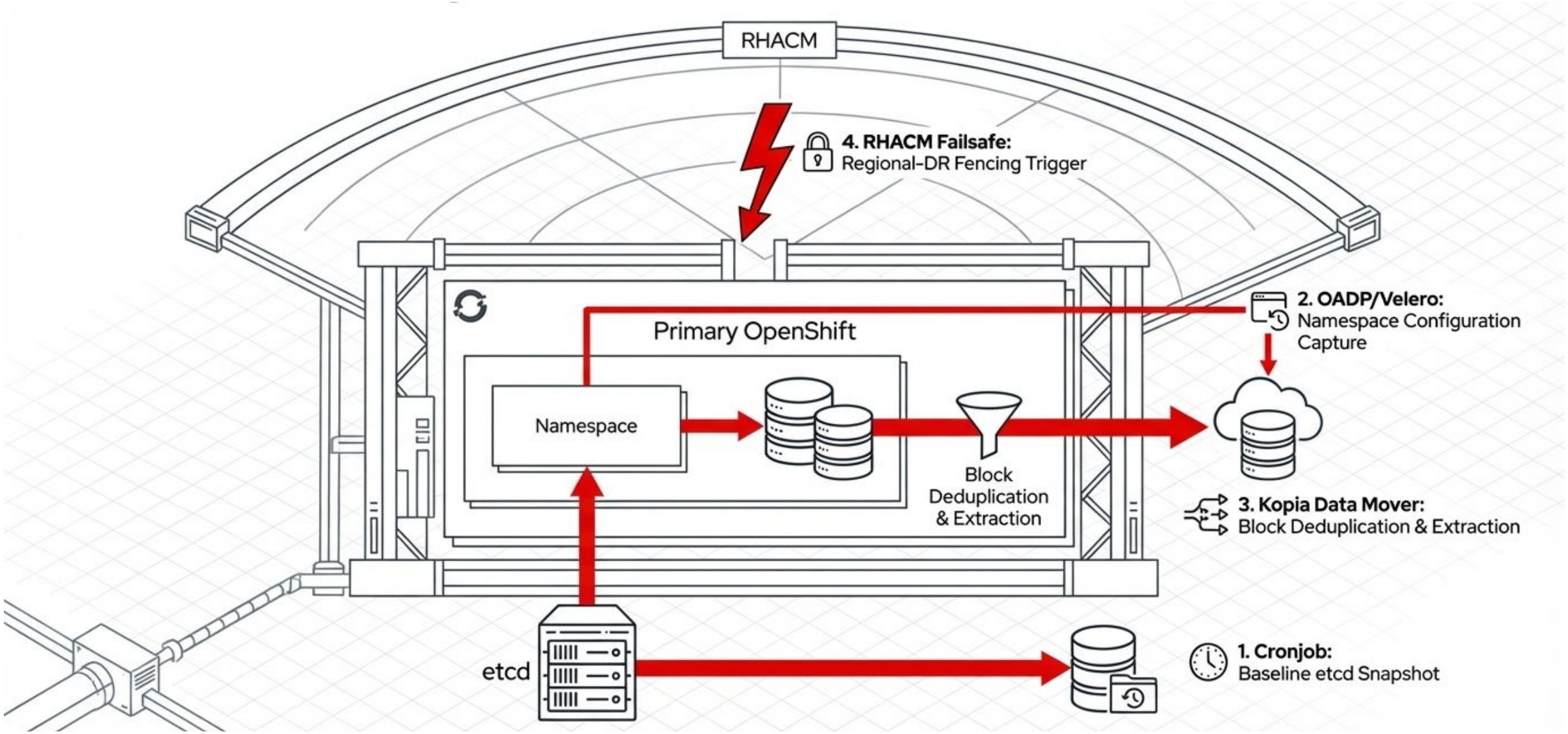


Outcome: Graceful shutdown. Synchronized data transfer. Zero data loss. No OS-level crash recovery required.

Emergency Failover



Outcome: Fencing' violently isolates the primary node to prevent split-brain data corruption. Workloads experience a hard halt. The OS must perform crash recovery upon reboot on the secondary site.


RHACM
4. RHACM Failsafe:
Regional-DR Fencing Trigger
Primary OpenShift
Namespace
Block
Deduplication
& Extraction
2. OADP/Velero:
Namespace Configuration
Capture
3. Kopia Data Mover:
Block Deduplication & Extraction
etcd
1. Cronjob:
Baseline etcd Snapshot

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DR – Planning Topics

- Identify RPO/RTO for hub vs. application clusters
- Network connectivity and DNS cutover between sites
- State: etcd, application data, and Git state (source of truth)
- Authentication: IdP availability during regional failure
- Compliance: where data may reside after failover

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Summary

- ACM provides multicluster management, policy, and application placement from a hub
- Hosted control plane patterns scale clusters; ACM manages them in the fleet
- GitOps integrates for consistent delivery across clusters and environments
- Resilience combines OpenShift design with hub-wide observability and policy
- DR requires storage, networking, and runbooks—ACM orchestrates parts of the story

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Thank You

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